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Paleomagnetic study of Cretaceous rocks from Pu'er, western Yunnan, China: Evidence of internal deformation of the Indochina block

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Abstract

Cretaceous red sandstones and siltstones were collected at 31 sties for paleomagnetic investigation from the Jingxing and Nanxin Formation around Pu'er (23.0°N, 101.0°E), western part of Yunnan province, China, in order to examine quantitative amount of tectonic deformation of the Simao Terrane due to collision of India with Asia. After stepwise thermal demagnetization, the high-temperature component magnetization with unblocking temperature of around 680 °C is isolated. Positive field tests show that the high-temperature component magnetization from 25 sites is primary for the Cretaceous formations. A tilt-corrected mean direction is $D=59.9°$, $I=45.2°$ with $\alpha_{95}=5.1°$, corresponding to a paleopole at 35.8°N, 173.1°E with $A_{95}=5.6°$. Comparison with Cretaceous paleomagnetic pole of the whole China block indicates that amount of clockwise rotation in the Pu'er area is evaluated to be 46±9° with respect to the whole China block, which is consistent with large amount of rotation in the Mengla and Jinggu areas (47° and 66°) in the central part of the Simao Terrane. Clockwise rotation of the central part is larger by more than 20° compared to the Yunlong and Yongping areas of northern part of the Simao Terrane, suggesting that the Simao Terrane experienced internal deformation. The southward displacement of the Pu'er area expected from inclination data is 7.5±6.6° with respect to the Chuan Dian fragment, which is a similar estimate to other areas in the Simao Terrane and the Indochina block. While the Simao Terrane displaced southward by 800 km together with clockwise rotation of 30° as a whole, further clockwise rotation occurred in the central part.
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Keywords: plate collision; deformation; Indochina block; Simao Terrane; Cretaceous; paleomagnetism

1. Introduction

The Indian continent collided with Asia at about 50 Ma and continued to penetrate into the Asian continent [1,2]. The 2500 km of convergence between India and Asia is partly accommodated in crustal

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thickening along with latitudinal shortening beneath the Tibetan Plateau and in lateral escape of blocks within Southeast Asia along major strike–slip zones [3–7]. Contribution to tectonic deformation of Southeast Asia provides constraints on the extrusion tectonics, which determines the timing and nature of extrusion of blocks.

Southeast Asia is a mosaic of tectonic blocks (e.g. the Simao Terrane, Qiangtang Terrane, Lhasa Terrane and the Yangtze craton: Fig. 1). The Simao Terrane, located in the northern part of the Indochina block that extends southwestward into Southeast Asia, is one of the prominent fragments of the extruded tectonic blocks. Because the Simao Terrane is situated between the stiff Indian continent and the Yangtze craton, the direct deformation feature of the Asian continent due to the penetration of India is expected to be preserved in this terrane.

Several paleomagnetic results from Cretaceous strata have been reported from the Simao Terrane in the last decade [8–12]. Paleomagnetic evidence on clockwise rotation of more than 25° has been provided from Cretaceous results. Southward displacement of around 10° in latitude is also suggested from the northern part of the Simao Terrane [8,12] as well as the Cretaceous paleomagnetic data from the Khorat Plateau in the central part of the Indochina block [13]. In the central part of the Simao Terrane, Cretaceous paleomagnetic data have been reported from three areas (Jinggu [10,11] and Mengla [10]). The fold tests they demonstrated were inconclusive [10], or were based on the sample level [11]. Their paleomagnetic declinations vary from 60° to 112°. Their paleomagnetic inclinations range from 36° to 43°, indicating insignificant southward displacement of the central part of the Simao Terrane. In order to validate these variable paleomagnetic directions, more reliable paleomagnetic data are needed for the central part of the Simao Terrane. This reliable paleomagnetic data offers key constraints to better understanding the complex deformation of the central part of the Simao Terrane.

The Pu'er area in the central part of the Simao Terrane, which is located between the Jinggu and Mengla areas, was chosen as a paleomagnetic study area. Late Mesozoic strata are widely exposed around the Pu'er area, allowing us to conduct more extensive paleomagnetic sampling and to demonstrate some field tests. Our paleomagnetic investigation focuses on the Cretaceous redbeds of the formations, because Cretaceous formations could have recorded evidence of Tertiary extrusion tectonics of the Indochina block due to the penetration of India since about 50 Ma.

2. Geological setting and sampling

The NW–SE trending Red River fault (Ailao Shan–Red River fault system) is a dominant tectonic feature of southern China. This is the boundary between the Chuan Dian fragment in the southwestern part of the Yangtze craton and two accretionary blocks (the Shan–Thai and Indochina blocks), and is connected with the Jinsha suture towards north in Tibet. Previous geologic and thermochronologic data [14,16–22] imply that a large amount of sinistral motion took place on the Red River fault in Tertiary time, whereas dextral motion occurred in Quaternary time [15].

The Shan–Thai and Indochina blocks, west of the Red River fault, consist of tectonic collages of continental fragments. The Baoshan Terrane and the Simao Terrane are well-defined continental fragments [23]. These two terranes are bounded by the Changning–Menglian suture [24,25]. The Baoshan Terrane, a part of the Shan–Thai block to the west of the Changning–Menglian suture, is clearly indicative of Gondwana affinities [24]. The Simao Terrane, a northern part of the Indochina block to the east of the Changning–Menglian suture, is of Cathaysian element [24]. These two terranes have amalgamated to form the Chuan Dian fragment until the Late Triassic [24].

Pu'er county (23.0°N, 101.0°E) is about 150 km west of the Red River fault and is situated in the Simao Basin in the central part of the Simao Terrane (Fig. 1b). Around Pu'er, Permo-Triassic to Cretaceous (locally Quaternary) cover strata are widely exposed. A detailed Cretaceous geology of the Simao Basin can be seen in the reports published by the Bureau of Geology and Mineral Resources of the Yunnan Province [26] and Leloup et al. [16]. The Cretaceous sequence of terrestrial origin is composed of four formations [26]; the Lower Cretaceous Jingxing Formation, the Middle Cretaceous Nanxin Formation, the Upper Cretaceous Hutousi Formation and the Mankuanhe Formation, in ascending order. The Lower Cretaceous Jingxing Formation, which is about 1300 m in thickness, mainly consists of gray to green feldspathic sandstones in the lower part and sandstone, siltstone and mudstone in the upper part. Lamellibranchiata such as *Estheria*, *Darwinula*, Gastropods and Sporopollen assign the age as Early Cretaceous to the Jingxing Formation [26]. The Middle Cretaceous Nanxin Formation, which is about 1450 m in thickness, mainly made up of purplish red sandstones. This formation conformably overlies the Lower Cretaceous Jingxing Formation and conformably underlies the Upper Cretaceous Hutousi Formation. Occurrence of *Estheria*, *Cypridea* (C), Charophyte and Sporopollen

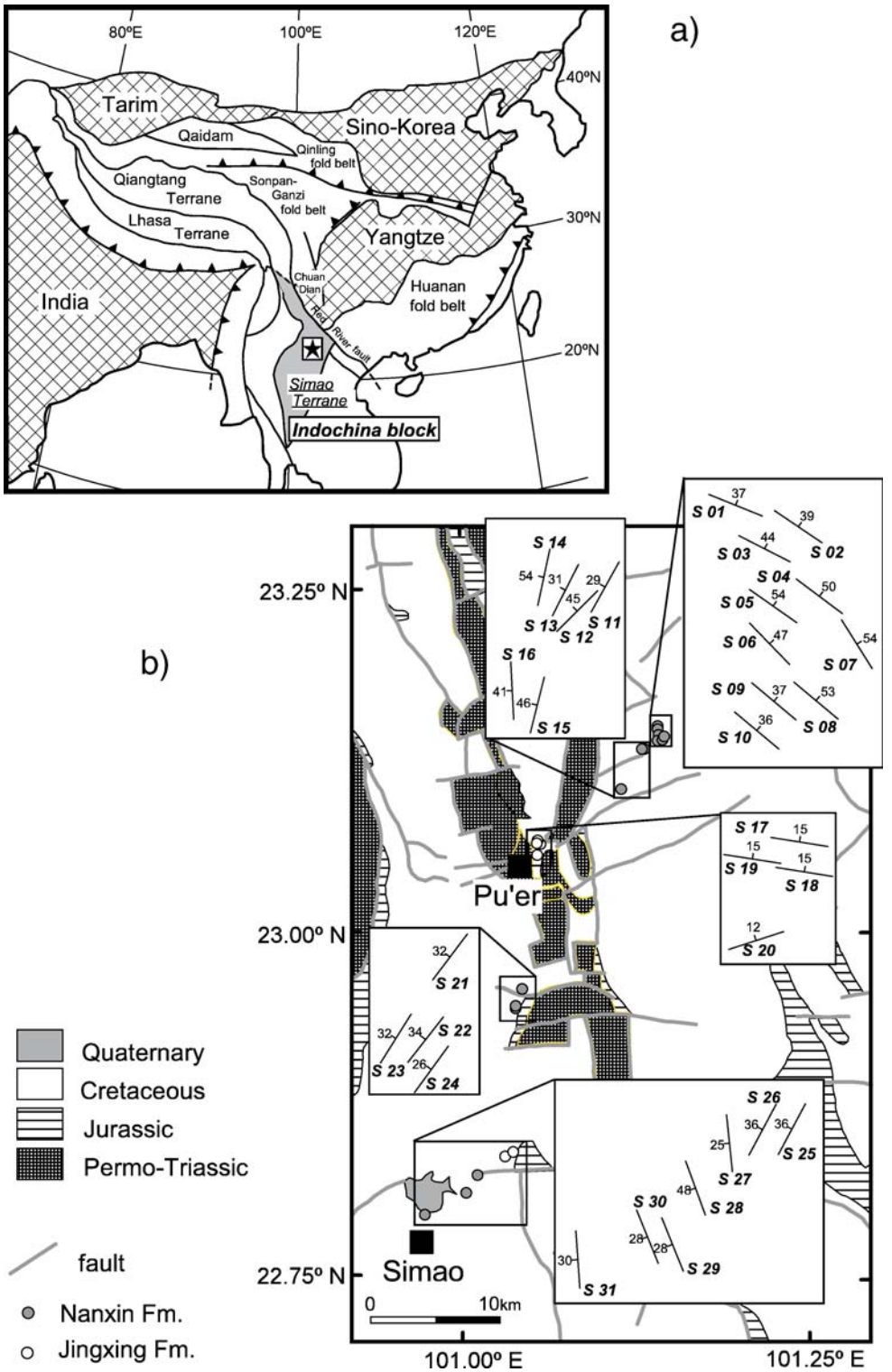


Fig. 1. (a) Sketch tectonic map of Southeast Asia (from Sato et al. [45]). The study area is marked with a star. Barbed and black lines are continental thrusts and suture zone, respectively. (b) Geological map of the study area (after [25,26]). Sampling sites are shown by open circles. Gray lines denote faults. Strike and dip of strata for each sampling site are shown in insets.

constraints the age as Middle Cretaceous to the Nanxin Formation [26].

Red sandstones and siltstones were collected at 31 sites (279 samples) in the Lower Cretaceous Jingxing Formation (63 samples from 7 sites) and the Middle Cretaceous Nanxin Formation (216 samples from 24 sites) along a road running from Simao county (22.84°N, 101.3°E) to northeast of Pu'er county (23.16°N, 101.14°E). The Cretaceous outcrops show several folding with a roughly NNW–SSE trending axis and dip from northeastward to westward at 12°–54°. The folding event probably occurred in the Late Eocene as indicated by the disconformity between the Middle–Upper Eocene continental molasses [16] and an unconformity between the Upper Eocene Denghei Formation and the Mengla Formation observed in the Simao Basin [26]. Nine hand samples, oriented with a magnetic compass, were distributed over a distance of up to 15 m at each site. The location of each sampling site was determined by a portable GPS. The present geomagnetic direction value at each sampling locality was estimated from the calculation using the International Geomagnetic Reference Field, 2000 [27].

3. Paleomagnetism

In the laboratory, one or more specimens, 25 mm in diameter and about 22 mm in height, were prepared from each sample. Remanent magnetization of each specimen was measured using a 2G Enterprises cryogenic magnetometer. Stepwise thermal demagnetization was carried out up to 690 °C with a Natsuhara TDS-1 thermal demagnetizer; the residual field in these furnaces was less than 5 nT. The magnetic behavior for each specimen during demagnetization was plotted on Zijderveld diagram [28]. Principal component analysis [29] was used to determine the component directions of magnetization, and a site mean direction was calculated using Fisherian statistics [30].

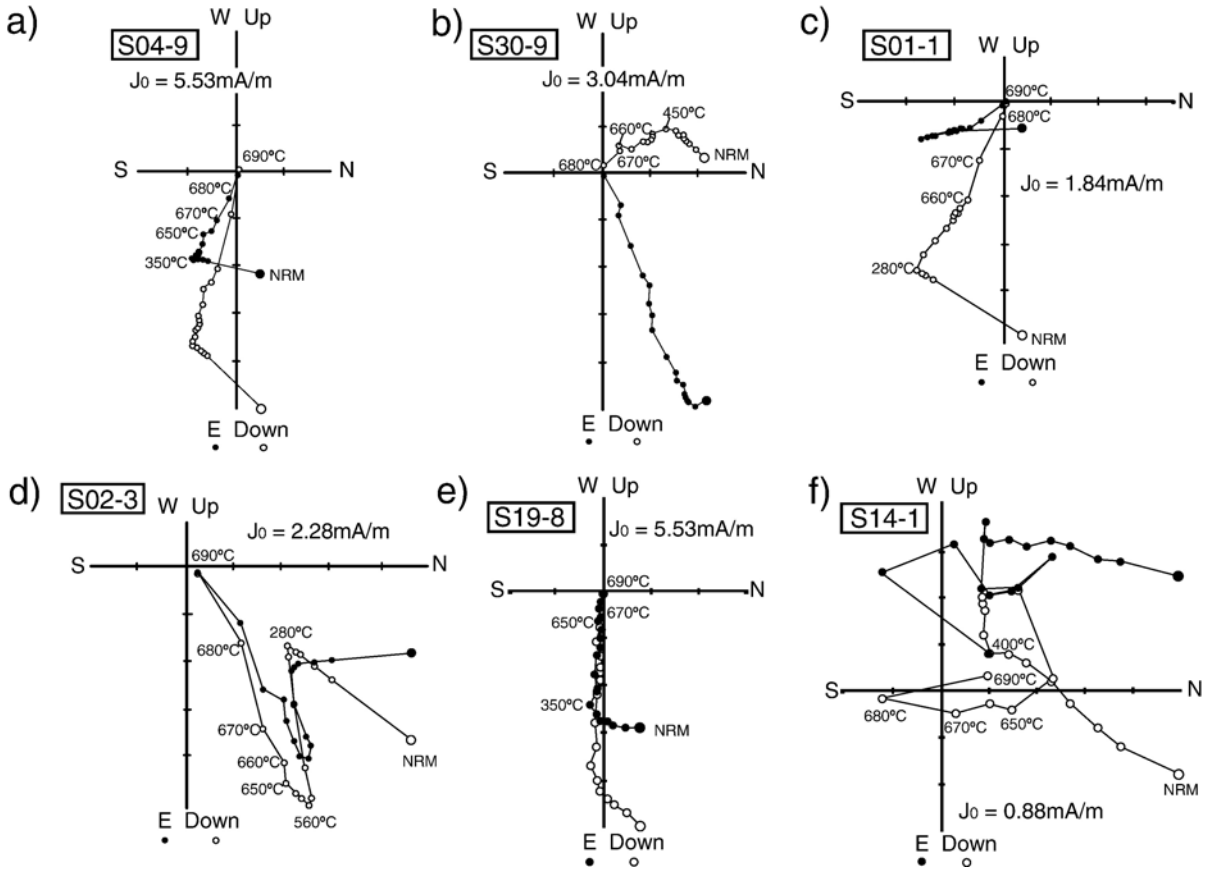


Fig. 2. Orthogonal projections of magnetization vector end-points for the representative specimens from the Lower Cretaceous Jingxing Formation (e) and the Middle Cretaceous Nanxin Formation (a–d, f) during thermal demagnetization experiments (in geographic coordinates). Solid symbols for horizontal projection, open symbols for vertical.

Progressive acquisition of isothermal remanent magnetization (IRM) up to 2.7 T and thermal demagnetization of composite IRMs [31] were performed on a 2G pulse magnetizer in order to identify ferromagnetic minerals. For composite IRMs, hard, medium and soft components were treated in DC fields of 2.7 T, 0.4 T, 0.12 T, respectively. The anisotropy of magnetic susceptibility (AMS) of some specimens was measured using a KLY-3 susceptibility bridge apparatus.

3.1. Demagnetization

Each specimen has natural remanent magnetization (NRM) intensity of 14.0–0.5 mA/m, and the majority of the NRM intensities ranges in 6.0–2.0 mA/m. All specimens of the Lower Cretaceous Jingxing Formation have a multi-component magnetization; generally two-component magnetization appears during progressive thermal demagnetization. The low-temperature component is removed by 350 °C. After removal of the low-temperature component, remanent magnetization decays linearly to the origin (Fig. 2e). The unblocking temperature of the high-temperature component ranges between 670 °C and 690 °C. High unblocking temperature above 670 °C indicates that this remanent magnetization is carried by hematite. Specimens of the Middle Cretaceous Nanxin Formation show similar thermal demagnetization behavior as specimens of the

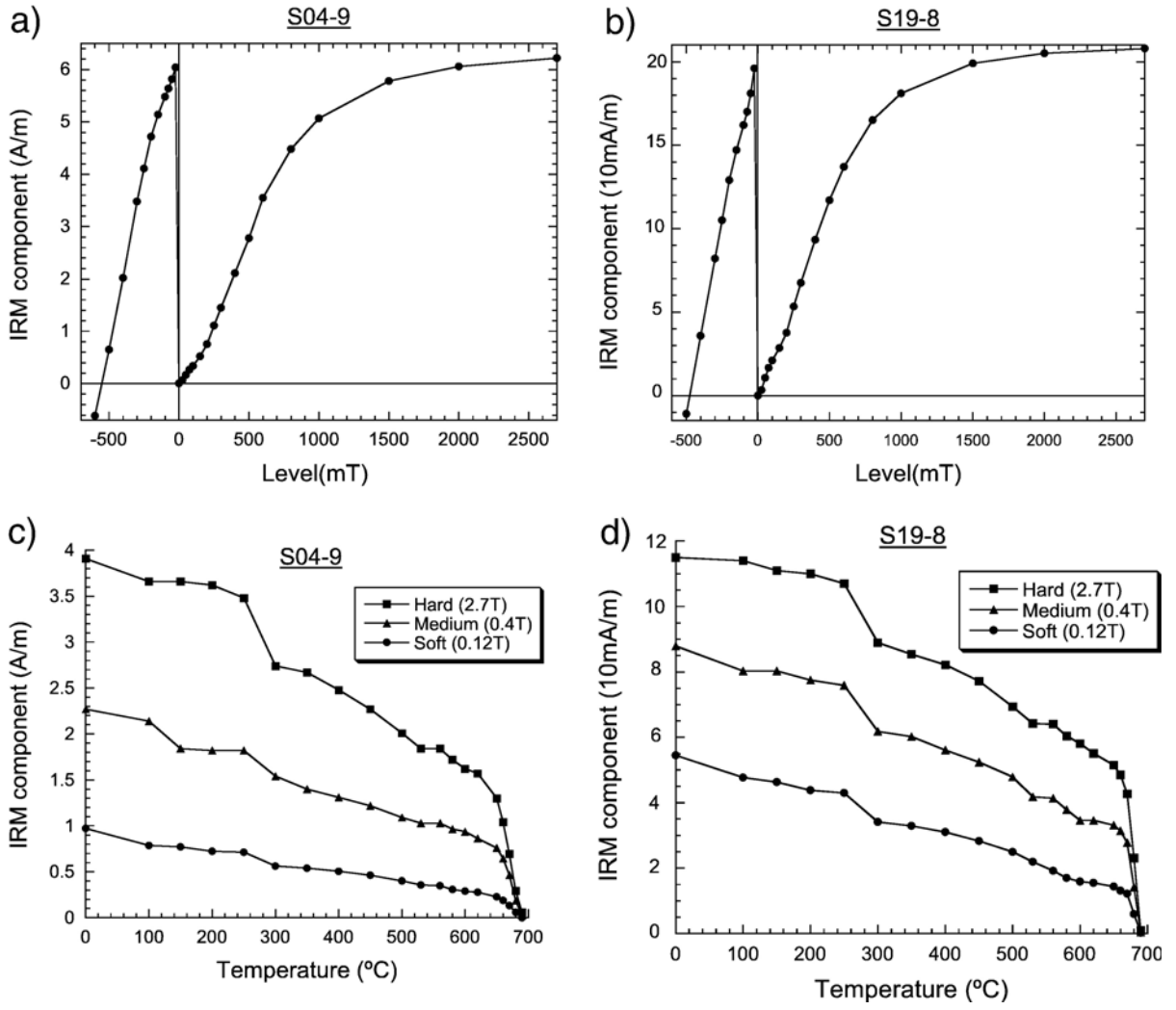


Fig. 3. (a, b) IRM acquisition curve and back-field demagnetization for the representative specimens (S04-9 and S19-8). Each specimen slightly increases and does not reach saturation IRM at field 2.7 T. (c, d) Thermal demagnetization of composite IRMs (2.7 T, 0.4 T and 0.12 T in DC field for three perpendicular axes of a sample) for the representative specimens (S04-9 and S19-8). Thermal demagnetization curves of all the components show the hematite unblocking temperature.

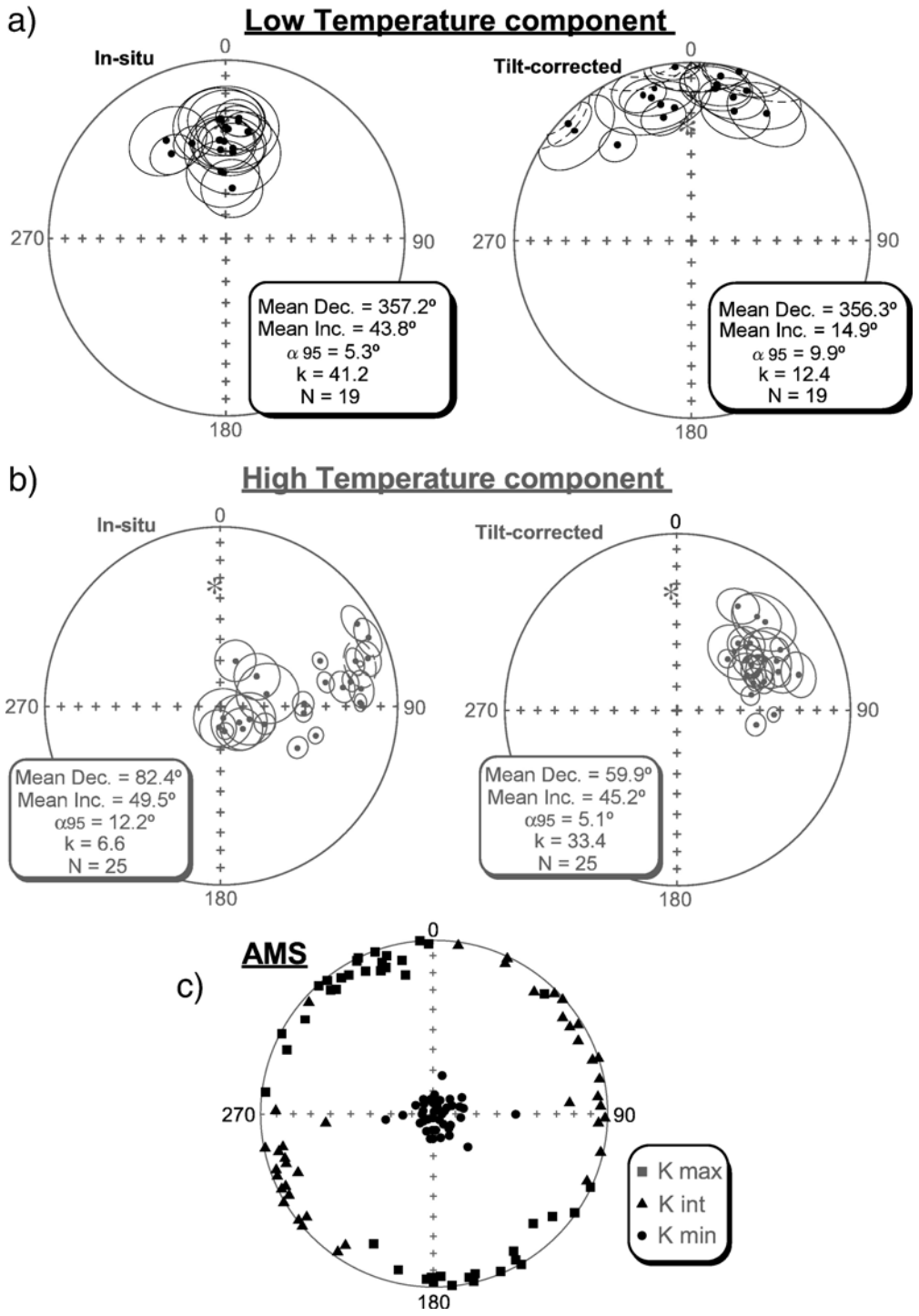


Fig. 4. (a) Equal-area projections of the site mean directions (circles) of low-temperature components with circles of 95% confidence limit before and after tilt correction. An asterisk represents the present direction of the Earth's magnetic field. The solid symbols refer to the lower hemisphere. (b) Equal-area projections of the site mean directions (circles) of high-temperature components with circles of 95% confidence limit before and after tilt correction. An asterisk represents the present direction of the Earth's magnetic field. The solid symbols refer to the lower hemisphere, whereas the open symbols refer to the upper hemisphere. (c) Equal-area lower projections of AMS principle axes after tilt correction for representative specimens of the Cretaceous formations.

Jingxing Formation. The low-temperature component is generally erased by 450 °C. The high-temperature component is unblocked between 670 °C and 690 °C (Fig. 2a–c). Only one specimen (S02-3 from site S02) has middle-temperature component magnetization between 280 °C and 560 °C (Fig. 2d). Specimens of sites S11, S12, S13, S14, S15 and S16, which have relatively weak NRM intensities of 1.5–0.5 mA/m, show unstable demagnetization behavior after the removal of the low-temperature component or from the initial stage of thermal demagnetization (Fig. 2f).

The IRM acquisition curve and back-field demagnetization of each specimen ensures a predominance of hematite as the magnetic carrier in the redbeds of the Cretaceous strata in the Pu'er section (Fig. 3a, b); the saturation field is more than 2.7 T and the coercivity of the remanent is 500–600 mT. Thermal demagnetization of composite IRMs suggests that unblocking temperature is around 680 °C in all the components (Fig. 3c, d). The high-temperature component in the Pu'er section is probably carried by hematite.

3.2. Paleomagnetic direction

For both formations, the low-temperature components were isolated from 19 sites (Fig. 4a, Table 1). The locality mean in geographic coordinate is $D=357.2°$, $I=43.8°$ ($k=41.2$, $\alpha_{95}=5.3°$), which is parallel to the present direction of the Earth's magnetic field ($D=358.7°$, $I=33.8°$). The precision parameter (k) for the 19 site mean directions decreases from 41.2 to 12.4 ($k_s/k_g=0.3$) after tilt correction. This suggests that this component has a post-folding origin due to viscous remanent magnetization acquired during the Bruhnes chron.

The high-temperature components were isolated from 26 sites at the Jingxing Formation (7 sites) and the Nanxin Formation (19 sites). Paleomagnetic directions display easterly deflection in declination, except for site S16 that shows westerly-deflected declination with upward inclination (Table 1). The direction of this site is not antipodal to the mean direction of the normal directions of other sites after tilt correction. Since the Middle Cretaceous Nanxin Formation including site S16, was formed during the Cretaceous Long Normal Superchron [32–34], the direction of this site is enigmatic. One possibility for the magnetization of this site is the timing of acquisition for this magnetization (e.g. excursion). An alternative explanation involves remagnetization, also observed in other areas of the Simao Terrane [8,11]. Because the direction of site S16 is hardly to be accepted as a Cretaceous paleomagnetic direction for the Pu'er area, the data of this site is therefore excluded from calculation of sites mean direction. The paleomagnetic directions of the remaining 25 sites display easterly deflection in declination and are well clustered at each site (Fig. 4b); precision parameter (k) ranges from 17.4 to 277.9. The mean direction of the 25 sites is $D=82.4°$, $I=49.5°$ with $\alpha_{95}=12.2°$ in situ and $D=59.9°$, $I=45.2°$ with $\alpha_{95}=5.1°$ after tilt correction. The precision parameter (k) for the overall site mean direction, increases monotonically from 6.6 at 0% to 33.4 at 100% unfolding. The optimal concentration of the direction–correction (DC) tilt test [35] is achieved at 91.6±12.9% untilting, which indicates a positive tilt test at 95% confidence level. According to the McFadden method [36], the fold test is also positive at 95% confidence level. The calculated value (ξ_1) is 20.44 in geographic coordinates and 5.672 after tilt correction, while the critical value (ξ_c) is 5.820. Thus the both Enkin- and McFadden-fold tests suggest that the high-temperature component was acquired before folding.

AMS results suggest that the samples from sites in the Pu'er area preserve an originally sedimentary fabric (Fig. 4c), although folding occurred in the Cretaceous sequence in the Simao Basin. Their magnetic fabrics are characterized by oblate susceptibility ellipsoid with maximum magnetic foliation of less than 3%. Axial distribution of AMS from these samples after tilt-correction shows that the k_{min} axes are generally vertical, whereas both of the k_{max} and k_{int} axes are dispersed in the horizontal plane. Non-deformed rocks characterize red sandstones and siltstones of the Jingxing and Nanxin Formations in the Pu'er section.

Consequently, the tilt-corrected mean direction of the high-temperature component is the characteristic direction of the Cretaceous formations, which corresponds to a paleomagnetic pole at 35.8°N, 173.1°E with $A_{95}=5.6°$ (Tables 1, 2).

4. Discussion

The Lower Cretaceous Jingxing Formation and the Middle Cretaceous Nanxin Formation in the Pu'er area preserve a reliable paleomagnetic direction: easterly-deflected declination is observed ($D=59.9°$, $I=45.2°$ with $\alpha_{95}=5.1°$). Reliability of the pre-folding nature for the remanent magnetization is ascertained through successful thermal demagnetization and positive fold tests. Characteristic directions for 25 sites after tilt correction is therefore regarded as primary magnetization acquired in the Cretaceous time.

Table 1
Paleomagnetic results for redbeds in the Cretaceous formations of the Simao Basin (Pu'er)

Site	Locality		Strike/dip		*n*/*N*	In-situ		Tilt-corrected		*k*	α_{95}	Formation
	(°N)	(°E)	(°)	(°)		Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)		(°)	
[High temperature component]												
Nanxin Fm.												
S01	23.16	101.14	292	36.5	9/9	113.1	75.7	44.6	51.5	19.2	12.1	Nanxin Fm.
S02	23.15	101.14	304	38.5	8/9	51.0	68.4	40.8	30.4	28.4	10.6	Nanxin Fm.
S03	23.15	101.14	297	44	8/9	74.7	67.8	45.2	29.2	17.4	13.7	Nanxin Fm.
S04	23.15	101.14	305	49.5	9/9	112.1	68.1	60.0	32.6	112.1	4.9	Nanxin Fm.
S05	23.15	101.14	307	54	9/9	168.0	78.6	47.9	43.0	204.8	3.6	Nanxin Fm.
S06	23.15	101.14	316	46.5	7/9	131.6	78.7	60.5	41.6	23.5	12.7	Nanxin Fm.
S07	23.15	101.14	328	54	8/9	162.1	84.5	64.0	37.0	18.1	13.4	Nanxin Fm.
S08	23.15	101.14	310	52.5	9/9	184.3	80.4	47.3	44.9	37.8	8.5	Nanxin Fm.
S09	23.15	101.14	310	37	5/9	19.1	68.1	30.6	31.7	72.8	9.0	Nanxin Fm.
S10	23.14	101.14	310	35.5	7/9	134.5	75.7	63.6	53.2	43.2	9.3	Nanxin Fm.
*S11	23.14	101.13	209	28.5	0/9							Nanxin Fm.
*S12	23.14	101.13	225	45	0/9							Nanxin Fm.
*S13	23.14	101.13	207	31	0/9							Nanxin Fm.
*S14	23.14	101.13	192	53.5	0/9							Nanxin Fm.
*S15	23.11	101.11	194	46	0/9							Nanxin Fm.
*S16	23.11	101.11	177	41	7/9	275.6	5.1	275.8	−35.4	45.6	9.0	Nanxin Fm.
S21	22.96	101.04	217	31.5	8/9	81.5	31.2	58.6	49.3	72.7	6.5	Nanxin Fm.
S22	22.95	101.04	218	33.5	9/9	79.4	26.6	57.9	45.0	156.3	4.1	Nanxin Fm.
S23	22.94	101.04	212	32	9/9	77.2	39.4	45.2	56.4	155.5	4.1	Nanxin Fm.
S24	22.94	101.04	216	26	8/9	66.1	38.4	42.9	47.2	277.9	3.3	Nanxin Fm.
S28	22.82	101.01	160	47.5	9/9	73.6	−19.0	73.9	28.4	32.7	9.1	Nanxin Fm.
S29	22.81	101.00	158	38	9/9	59.2	11.9	55.2	49.3	56.6	6.9	Nanxin Fm.
S30	22.81	101.00	158	38	9/9	65.5	9.4	64.7	47.2	50.9	7.3	Nanxin Fm.
S31	22.80	100.97	176	30	9/9	74.0	21.5	65.8	50.3	97.5	5.2	Nanxin Fm.
Formation mean direction					18/24	77.3	54.6			5.6	16.1	
					18/24			53.7	43.2	43.5	5.3	
Jingxing Fm.												
S17	23.07	101.06	279	15	9/9	118.2	49.3	100.0	52.2	111.2	4.9	Jingxing Fm.
S18	23.07	101.05	279	15	9/9	106.7	43.6	92.3	43.9	251.5	3.3	Jingxing Fm.
S19	23.07	101.06	279	15	9/9	87.8	51.3	71.4	46.5	102.8	5.1	Jingxing Fm.
S20	23.07	101.05	252	12	9/9	93.7	51.4	78.3	54.5	142.2	4.3	Jingxing Fm.
S25	22.84	101.04	208	36	9/9	82.5	21.6	64.5	48.3	54.3	7.1	Jingxing Fm.
S26	22.84	101.04	208	36	9/9	87.8	22.9	70.1	51.8	214.0	3.5	Jingxing Fm.
S27	22.84	101.03	174	25	9/9	71.9	14.4	69.1	38.7	62.2	6.6	Jingxing Fm.
Formation mean direction					7/7	90.9	37.3			16.3	15.4	
					7/7			77.6	48.6	61.6	7.7	
Overall mean direction (Nanxin + Jingxing)					25/31	82.4	49.5			6.6	12.2	
					25/31			59.9	45.2	33.4	5.1	
[Low temperature component]					19/31	357.2	43.8			41.2	5.3	
					19/31			356.3	14.9	12.4	9.9	

The magnetization directions of high temperature components are listed. *N* and *n* are numbers of field oriented samples and samples used for statistics, respectively. Dec. and Inc. are declination and inclination, respectively; *k* is the Fisherian precision parameter [30]; α_{95} is the radius of cone of 95% confidence about the mean direction. *S11, S12, S13, S14, S15 and S16 are not used for calculation of the mean direction for the high temperature component.

Table 2
Cretaceous paleomagnetic results obtained from the Indochina block and its surrounding regions

	Locality			Age	N	Observed direction			VGP			Rotation	Latitudinal difference		Reference
	Name	Lat. (°N)	Lon. (°E)			Dec. (°)	Inc. (°)	α_{95} (°)	Lat. (°N)	Lon. (°E)	A_{95} (°)	(°)	(°)	Relative to	
Indochina block															
Simao Terrane:															
	Pu'er	23.0	101.0	K1–2	25	59.9	45.2	5.1	35.8	173.1	5.6	46.3±8.9	9.0±6.7	A	This study
	Lanping	26.5	99.3	E	9	266.1	−39.8	11.2	14.5	169.7	10.9	90.7±15.4	8.6±11.5	B	[45]
	Yunlong	25.8	99.4	K2	20	40.2	49.9	3.9	54.6	171.3	4.4	26.6±7.6	10.7±5.7	A	[8]
	Yunlong	25.8	99.4	K2	29	38.3	50.7	3.4	56.7	170.1	4.0	24.6±7.1	11.4±5.4	A	[12]
	Xiaguan	25.6	100.2	K2	9	6.9	47.7	8.6	83.6	152.7	10.0	−6.7±13.6	8.8±10.7	A	[10]
	Yongping	25.5	99.5	K1	12	42.0	51.1	15.7	50.9	167.3	20.6	28.4±25.9	10.7±8.3*	A	[9]
	Jinggu	23.4	100.4	K1	–	84.4	39.6	17.8	13.6	171.5	–	70.8±23.8	4.5±18.2	A	[11]
	Jinggu	23.4	100.5	K2	–	295.8	−36.0	6.3	−13.9	161.3	–	102.2±9.0	2.0±7.3	A	[11]
	Jinggu	23.5	100.8	E	–	84.7	38.9	7.6	13.2	172.2	–	89.1±10.9	11.0±6.8	B	[11]
	Jinggu	23.4	100.9	K2	8	79.4	43.3	9.1	18.9	170.0	8.9	65.8±13.3	7.1±9.6	A	[10]
	Mengla	21.6	100.4	K2	10	60.8	37.8	7.6	33.7	179.3	8.2	47.4±10.5	5.0±9.0	A	[10]
Khorat plateau:															
	Khon Kaen	16.5	103.0	K1	10	28.1	40.5	2.4	62.7	173.3	2.4	14.7±5.1	11.3±4.4	A	[13]
Stable part of the China block															
A	Whole China block			K1–2	10 areas				76.2	211.2	3.7				[37]
Yangtze craton															
	Ya'an	30.0	102.9	Ju-P	26	2.0	34.2	3.6	78.6	273.4	2.9				[38]
B	Ya'an	30.0	103.0	P	7	356.0	31.9	4.1	76.8	299.9	3.6				[38]
	Xichang	27.9	102.3	K1–2	33	3.7	41.5	3.4	85.2	241.7	3.5				[39]
	Huili	26.6	102.3	K2	18	8.1	38.8	6.6	81.5	220.9	7.1				[40]
	Yuanmou	25.9	101.8	K2	21	26.9	35.6	3.6	64.6	199.6	3.3				[41]
	Yongren	26.1	101.7	P–E	16	17.2	26.6	5.8	70.1	224.6	4.9				[46]
	Dayao	25.7	101.3	P–E	10	16.5	31.1	4.8	72.3	218.4	4.5				[46]
	Chuxiong	25.0	101.5	K1	21	44.6	41.3	10.7	49.2	182.0	11.4				[42]
Song Da Terrane															
	Lai Chau	22.3	103.4	K2	5	12.2	40.1	4.7	78.7	188.0	5.1				[47]
	Yen Chau	21.0	104.4	K2	8	3.2	26.7	12.9	83.2	255.6	10.8				[47]
Qiangtang Terrane															
	Markam	29.7	98.7	K2	11	39.7	51.6	8.5	56.7	172.7	10.6	25.8±14.6	8.5±11.2	A	[42]
	Markam	29.7	98.4	K1	12	57.7	50.2	10.9	40.6	170.5	13.0	44.8±17.9	7.4±13.5	A	[42]

Lat. and Lon. are latitude and longitude, respectively. Age: Ju, Upper Jurassic; K1, Lower Cretaceous; K2, Upper Cretaceous; P, Paleocene; E, Eocene. N is the number of sites used for statistics. Dec. and Inc. are declination and inclination, respectively. α_{95} and A_{95} are the radius of cone of 95% confidence about the mean direction and the VGP, respectively. Rotational and latitudinal difference are evaluated by comparison between the observed paleomagnetic declination and that of calculated from the Cretaceous paleomagnetic pole of whole China block [36] and Tertiary paleomagnetic pole of Ya'an [38]. Latitudinal difference of Yongping (*) is evaluated using recalculated inclination data (bivariate mean) by Yang et al. [44].

4.1. Comparison with other Cretaceous paleomagnetic data

4.1.1. Comparison with Cretaceous paleomagnetic pole of the whole China block

The Pu'er pole is compared with the Cretaceous paleomagnetic pole of the whole China block (from the Sino–Korea craton, the Yangtze craton and the Huanan fold belt: the North and South China block) calculated by Yang and Besse [37]. The Pu'er pole is significantly distinguished from the Cretaceous pole of the whole China block (Fig. 5a). Angular distance between the Pu'er pole and the whole China block is 43.9±6.7°. This indicates that the Pu'er area was subjected to tectonic

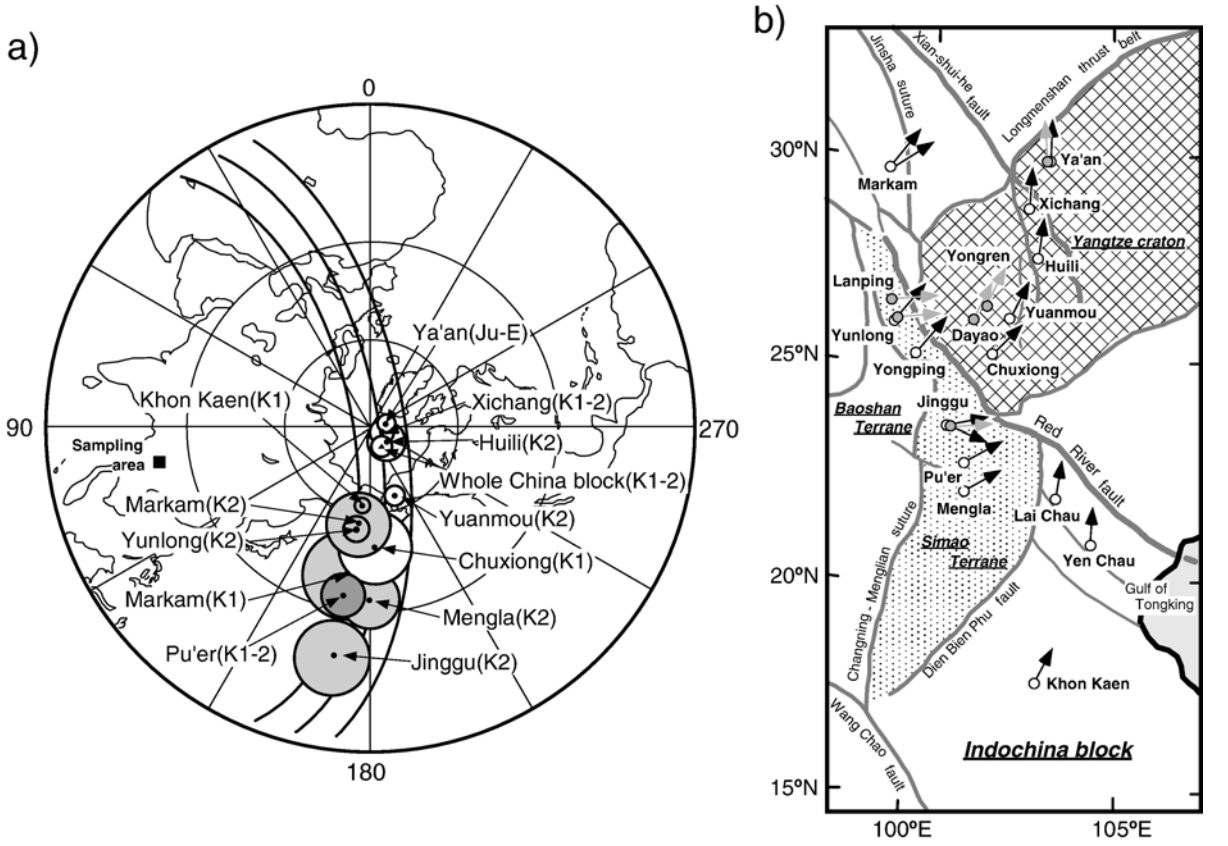


Fig. 5. (a) Cretaceous paleomagnetic poles of twelve areas around the Red River fault and whole China block calculated by Yang and Besse [37] with associated circles of 95% confidence. Small circles centered on our study area are also shown. Shaded poles are from west of the Red River fault (the Qiangtang Terrane and the Indochina block) and open poles are from east of the Red River fault. Ju, Upper Jurassic; K1, Lower Cretaceous; K2, Upper Cretaceous; E, Paleogene. (b) Declinations of the Cretaceous (black) and Tertiary (gray) paleomagnetic directions of the Indochina block and its surrounding regions are shown by arrows.

movement with respect to the whole China block since Cretaceous time.

4.1.2. Differential clockwise rotation within the Simao Terrane

One of the prominent tectonic phenomena for the Pu'er area is a large clockwise rotation with respect to the whole China block. The amount of rotation is evaluated from comparison between observed and expected paleomagnetic directions calculated from the pole of the whole China block at Pu'er: the Pu'er area is rotated clockwise through 46.3±8.9°.

The large amount of clockwise rotation in the central part of the Simao Terrane expected from previous paleomagnetic data is confirmed by our reliable paleomagnetic data from Pu'er. Various amount of clockwise tectonic rotation is expected in the central part of the Simao Terrane [10,11]. Easterly declinations are reported from the Mengla area (21.6°N, 100.4°E; D=61°) of the Middle Cretaceous Mangang Formation in central part of the Simao Terrane [10]. Paleomagnetic results from Cretaceous strata in the Jinggu area (23.5°N, 100.6°E) have been reported by Chen et al. [11]. Their paleomagnetic directions of the Jingxing Formation (D=84°) are consistent with the paleomagnetic data of Huang and Opdyke (23.4°N, 100.9°E; D=79°) [10]. Paleomagnetic data of Chen et al. [11] from the Middle Cretaceous Mangang Formation, in contrast, show large amount of clockwise rotation (more than 100°). The central part of the Simao Terrane is characterized by large clockwise rotation of more than 46°.

In contrast to the central part, about 30° of clockwise rotation is observed in northern part of the Simao Terrane. Paleomagnetic data from Yunlong (25.8°N, 99.4°E) in the central part of the Lanping Basin, which is in the northern part of the Simao Terrane [8,12], is rotated clockwise through 25°–27° with respect to the whole China block. About 28° of clockwise tectonic rotation is

also observed from Yongping [9] (25.5°N, 99.5°E) in the southern part of the Lanping Basin, indicating that both the Yongping and Yunlong areas in the northern part of the Simao Terrane have behaved as a rigid body.

Large clockwise rotation observed in limited areas of the central part of the Simao Terrane is described as differential tectonic rotation within this terrane. This differential rotation is ascribable to internal deformation due to the soft and deformable nature of the Simao Terrane that flanks the stiff Yangtze craton and the central part of the Indochina block. The deformed zone extends more than 1000 km along the Red River fault; from the eastern Qiangtang Terrane (30°N, 99°E, [43]) to the central part of the Indochina block (Khorat Plateau; 16°N, 103°E, [13]). The paleomagnetic poles obtained from the Qiangtang Terrane and the Indochina block, west of the Red River fault, are streaked out along a small circle centered on our study area (Fig. 5a). This indicates that the differential clockwise rotations, including local rotations, took place for these areas. Since existence of the fairly long deformation zone is observed throughout Southeast Asia to the northeast of India from paleomagnetic data [8–13,42], the clockwise rotation observed in the deformed zone is ascribed to tectonic deformation due to collision of India. The deformed zone experienced variable rotation rather than uniform rotation (Fig. 5b). Amounts of rotation in the central part of the Simao Terrane are quite larger (more than 18°) than that of the Lanping Basin in the northern part of the Simao Terrane. Paleomagnetic data near Khon Kaen (Khorat Plateau; 16.5°N, 103.0°E) in the central part of the Indochina block [13] show clockwise rotation of about 15° with respect to the whole China block, indicating that the Khorat Plateau in the central part of the Indochina block is relatively rigid and experienced small amount of clockwise rotation with respect to the Simao Terrane.

4.1.3. Southward extrusion of the Indochina block

Latitudinal displacement is claimed for the terranes west of the whole China block from comparison of the paleomagnetic poles. The amount of southward displacement is estimated for the Pu'er area with respect to the whole China block. The paleolatitude of the Pu'er area calculated from the observed inclination is 26.7±5.6°N, whereas the paleolatitude of the Pu'er area expected from the whole China block is 17.7±3.7°N [37]. The amount of latitudinal difference between the observed paleolatitude and the expected paleolatitude is 9.0±6.7°, corresponding to 1000±750 km (Fig. 6). Paleomagnetic data of the Pu'er area gives further evidence that the central part of the Simao Terrane has experienced southward displacement since Cretaceous time.

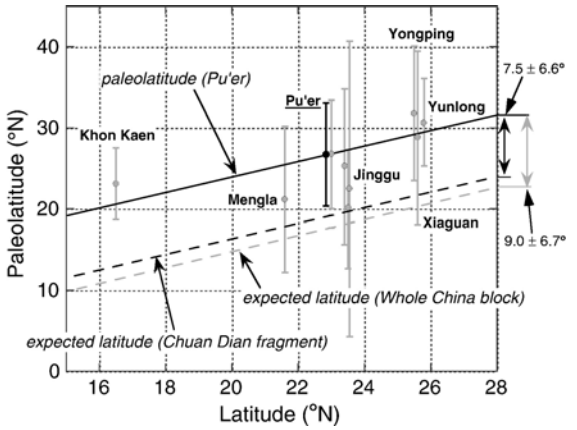


Fig. 6. Plots of paleolatitude of Pu'er [This study], Yunlong [8], Xiaguan [10], Yongping [9], Jinggu [10,11], Mengla [10] and Khon Kaen [13] for Cretaceous time. Black plot that represents the latitudinal difference between the data of Pu'er and the data of Xichian [39] is also shown. Paleolatitude and its uncertainty are calculated after Butler [54]. The error bar shows uncertainty at 95% confidence level.

Southward displacement of the Indochina block is associated with southeastward extrusion of the Indochina block along the Ailao Shan fault, northern part of the Red River fault, due to penetration of India into Asia (e.g. [8,12,13,16–20,44]). More rigorous amount of southward motion of the Simao Terrane along the Ailao Shan fault is calculated by comparison of the Cretaceous paleomagnetic data of Pu'er with recently reported reliable paleomagnetic data of Xichang [39] from the Chuan Dian fragment in the southwestern margin of the Yangtze craton. Tamai et al. [39] estimated more than 350 km of southward displacement of the Chuan Dian fragment with respect to the Sichuan Basin between the Ailao Shan fault and the Xianshuihe–Xiaojiang fault system since Cretaceous time. Thus, this comparison gives a more precise estimation of the southward motion of the Simao Terrane associated with sinistral motion of the Ailao Shan fault, comparing with that of the whole China block [37]. The estimated amount of southward displacement of the Pu'er area with respect to the Chuan Dian fragment is 7.5±6.6° (corresponding to 800±700 km: Fig. 6). This southward motion of the Indochina is conversed to southeastward motion by the present trend of the Red River fault (140°); 1040±900 km of southeastward motion of the Simao Terrane along the Ailao Shan fault is evaluated. Geological marker and thermochronologic data show that 700±200 km of sinistral motion along the Red River fault occurred between 35 Ma and 17 Ma [15–17,19–22]. The estimated amount of southeastward motion from a paleomagnetic view-point is greater (although compatible within 95% confidence limit) than the estimated

amount of southeastward displacement from geologic and thermochronologic data. These indicate that southeastward motion of the Indochina block can be explained mainly by sinistral motion of the Ailao Shan fault [16,17,20,21]. The remaining amount of this motion might have been absorbed by major strike–slip faults around the Simao Terrane (e.g. the Gaoligong fault, the Dien Bien Phu fault and the Changning–Menglian suture), and by internal deformation associated with differential clockwise rotation due to motion of subsidiary existing strike–slip faults within the Simao Terrane [22].

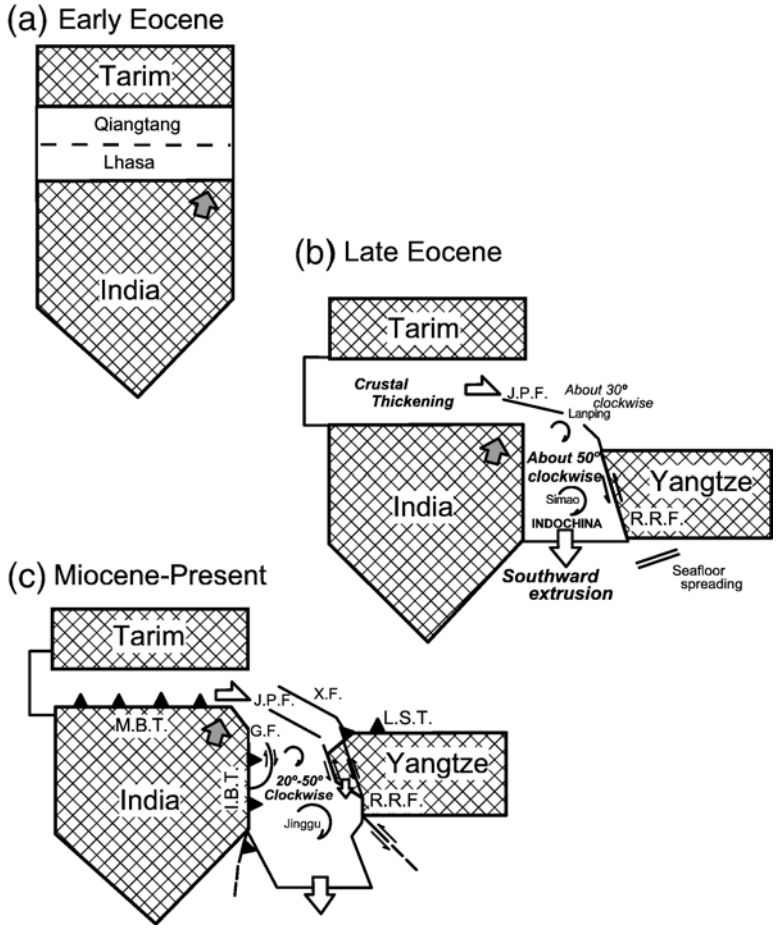


Fig. 7. A synthetic tectonic model on the basis of paleomagnetic, geological and thermochronological evidence. (a) After India collided with Asia at the Early Eocene, India began to penetrate into Asia. (b) Tectonic deformation of the Asian continent commenced since the Late Eocene. The indentation of India was partly absorbed by thickening of the Tibetan Plateau and the Himalayas [5]. In the Indochina block east of India, the penetration of India was accommodated by clockwise rotational motion and southward displacement. The central part of the Simao Terrane (Simao Basin) has been subject to clockwise rotation of about 50° and initiated southward displacement of about 800 km since this time until present. The northern part of the Simao Terrane (Lanping Basin) has experienced clockwise rotational motion of about 30° and started southward displacement of more than 1000 km since this time until present [8,9,12]. These large displacements were associated with left-lateral motion of the Ailao Shan fault northern part of the Red river fault (R. R. F.) between 35 Ma and 17 Ma and opening of the South China Sea. Open arrows indicate direction of displacement of terranes. J. P. F.=Jiali–Parlung fault. (c) Continental deformation of the Asia has continued until Miocene to Pliocene, and possibly until present. Thrusting at the following thrusts started at around 20 Ma: the Main Boundary thrust (M. B. T.) of the Himalayas, Indo–Burman thrust (I. B. T.) of the Indo–Burman Range and the Longmenshan thrust (L. S. T.) of the Longmen Mountains. In this time the Chuan Dian fragment (C. D. F.), southwestern margin of the Yangtze craton, started southward displacement with respect to the Sichuan Basin accompanied by the motion of the Indochina block between the Xianshuihe–Xiaojiang fault system (X. F) and the Ailao Shan fault. Since Miocene, eastward thrusting and décollements have occurred in this region since Miocene [20]. The Eocene strata in the Lanping Basin, might have suffered additional differential clockwise rotation of 50° since this time until present [45], possibly associated with these eastward thrusting and décollements occurred in this region [20]. The Cretaceous strata of the Jinggu area in the Simao Basin possibly have been also subjected to additional local clockwise rotational motion by these thrusting and décollements until present [10,11]. G. F.=Gaoligong fault.

4.2. Synthetic tectonic model

A realistic tectonic model is proposed on the basis of paleomagnetic, geological and thermochronological evidence (Fig. 7). This model explains more precise deformation features of the Simao Terrane than those previously advocated by Sato et al. [8]:

(a) After India collided with Asia in the Early Eocene, India began to penetrate into Asia.
(b) Tectonic deformation of the Asian continent commenced in the Late Eocene. The indentation of India was partly absorbed by thickening of the Tibetan Plateau and thrusting in the Himalayas [5]. In the Indochina block to the east of India, the penetration of India was accommodated by clockwise rotational motion and southward displacement. The most remarkable point is that the magnitude of clockwise rotation is different between northern and central parts of the Simao Terrane. The central part of the Simao Terrane (Simao Basin) has been subjected to clockwise rotation of about 50° and underwent southward displacement of 800 km from Late Eocene until present. The northern part of the Simao Terrane (Lanping Basin) has experienced clockwise rotational motion of about 30° and southward displacement of more than 1000 km from Late Eocene until present [8,9,12]. These large displacements were associated with left-lateral motion of the Ailao Shan fault between 35 Ma and 17 Ma, and with seafloor spreading of the South China Sea [16–18,48].
(c) Continental deformation of Asia continued until Miocene to Pliocene, and possibly until present. The Miocene penetration of India was absorbed by thrusting in the Himalayas, the Indo–Burman Range and the Longmen Mountains started at around 20 Ma [19,49,50]. The central part of the Indochina block (Khorat Plateau) is relatively rigid with respect to the Simao Terrane, and has been experiencing clockwise rotation of about 15° until present [13,44]. In this time the Chuan Dian fragment in the southwestern margin of the Yangtze craton, started southward displacement with respect to the Sichuan Basin accompanied by the motion of the Indochina block between the Xianshuihe–Xiaojiang fault system and the Ailao Shan fault [22,51]. Since Miocene, the deformation phase (shortening phase) has changed from southward to eastward on the surface strata of the Simao Terrane by the development of the Indo–Burman Range. Based on geological observation, eastward thrusting and décollements have occurred in this region [20]. The Eocene strata in the Lanping Basin, might have suffered additional differential clockwise rotation of 50° since deposition [45], possibly associated with these eastward thrusting and décollements occurred in the region [20]. The Cretaceous strata of the Jinggu area in the Simao Basin have also possibly been subjected to additional local clockwise rotational motion by these thrusting and décollements until present [10,11]. GPS measurements indicate that the South China regions including the Indochina block and the Yangtze craton are now moving south–southeastward at an average velocity of 6–11 mm/yr with respect to the stable Eurasia [52,53]. Complicated crustal deformation observed in the Simao Terrane is a reflection of intensive tectonic deformation of this region due to the collision of India with Asia. This tectonic deformation might be superimposed complicatedly by the southeastward extrusion of Indochina since the Late Eocene, and eastward thrusting and décollements due to the development of the Indo–Burman Range from the Miocene until present [8–12,14–22,52,53].

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